CALCULUS FOR ML FOUNDATIONS

Derivative Rules Quiz

Twenty-two questions on the standard rules of differentiation — the same rules that power gradients and backpropagation. Work through them, then submit to see your score and worked explanations.

⏱ 30 minutes | 22 questions | 1 mark each | Auto-submit at 0:00

Question 1 of 22

What is the derivative of a constant, `d/dx (c)`?

- ◯ `0` — correct
- ◯ `1`
- ◯ `c`
- ◯ `x`

Correct. The constant rule: a constant never changes, so its rate of change is `0`.

Question 2 of 22

Using the power rule, `d/dx (x⁵)` = ?

- ◯ $5x^4$ — correct
- ◯ x^4
- ◯ $5x^6$

- ◯ $4x^5$

Correct. Power rule: $d/dx\ (x^n) = n \cdot x^{n-1}$. Here n = 5, giving $5x^4$.

Question 3 of 22

What is $d/dx\ (7x^3)$?

- ◯ $21x^2$ correct
- ◯ $7x^2$
- ◯ $21x^3$
- ◯ $3x^2$

Correct. The constant multiple stays: $7 \cdot 3x^2 = 21x^2$.

Question 4 of 22

Apply the sum rule: $d/dx\ (3x^2 + 2x - 5)$ = ?

- ◯ $6x + 2$ correct
- ◯ $6x + 2 - 5$
- ◯ $3x + 2$
- ◯ $6x^2 + 2$

Correct. Differentiate term by term: $6x$, 2, and the constant $-5 \to 0$. Result: $6x + 2$.

Question 5 of 22

What is $d/dx\ (\sqrt{x})$?

- ◯ `1 / (2√x)` — correct
- ◯ `2√x`
- ◯ `√x / 2`
- ◯ `1 / √x`

Correct. Write `√x = x^(1/2)`. Power rule gives `½·x^(−1/2) = 1/(2√x)`.

Question 6 of 22

What is `d/dx (sin x)`?

- ◯ `cos x` — correct
- ◯ `−cos x`
- ◯ `−sin x`
- ◯ `sin x`

Correct. A standard result: the derivative of `sin x` is `cos x`.

Question 7 of 22

What is `d/dx (cos x)`?

- ◯ `−sin x` — correct
- ◯ `sin x`
- ◯ `−cos x`
- ◯ `cos x`

Correct. The derivative of `cos x` is `−sin x` (note the minus sign).

Question 8 of 22

What is `d/dx (eˣ)`?

- ◯ e^x — correct
- ◯ $x \cdot e^{x-1}$
- ◯ e^{x-1}
- ◯ $x \cdot e^x$

Correct. The exponential e^x is its own derivative — one reason it appears so often in ML.

Question 9 of 22

What is `d/dx (ln x)`?

- ◯ `1 / x` — correct
- ◯ `1 / (x ln x)`
- ◯ `ln x`
- ◯ `x`

Correct. The natural logarithm differentiates to $1/x$ (for $x > 0$).

Question 10 of 22

What is `d/dx (1/x)`?

- ◯ $-1 / x^2$ — correct

- ◯ $1 / x^2$
- ◯ $-1 / x$
- ◯ $\ln x$

Correct. Write $1/x = x^{-1}$. Power rule: $-1 \cdot x^{-2} = -1/x^2$.

Question 11 of 22

Which rule do you use to differentiate a product like $x^2 \cdot \sin x$?

- ◯ Product rule — correct
- ◯ Chain rule
- ◯ Quotient rule
- ◯ Power rule

Correct. Two functions multiplied together call for the product rule: $(uv)' = u'v + uv'$.

Question 12 of 22

Using the product rule, $d/dx\ (x^2 \cdot \sin x)$ = ?

- ◯ $2x \cdot \sin x + x^2 \cdot \cos x$ — correct
- ◯ $2x \cdot \cos x$
- ◯ $x^2 \cdot \cos x$
- ◯ $2x \cdot \sin x - x^2 \cdot \cos x$

Correct. (uv)′ = u′v + uv′ with u = x², v = sin x gives 2x·sin x + x²·cos x.

Question 13 of 22

Using the chain rule, d/dx (3x + 1)⁴ = ?

- ◯ 12(3x + 1)³ — correct
- ◯ 4(3x + 1)³
- ◯ 12(3x + 1)⁴
- ◯ 3(3x + 1)³

Correct. Outer power: 4(3x+1)³, times the inner derivative 3 → 12(3x+1)³.

Question 14 of 22

Using the chain rule, d/dx (e^(2x)) = ?

- ◯ 2·e^(2x) — correct
- ◯ e^(2x)
- ◯ 2x·e^(2x)
- ◯ e^(2x)/2

Correct. Derivative of the outer eᵘ is eᵘ; times inner derivative 2 → 2e^(2x).

Question 15 of 22

Using the chain rule, d/dx (sin(x²)) = ?

- ◯ 2x·cos(x²) — correct

- ◯ $\cos(x^2)$
- ◯ $2x \cdot \sin(x^2)$
- ◯ $\cos(2x)$

Correct. Outer: $\cos(x^2)$; inner derivative of x^2 is $2x$ → $2x \cdot \cos(x^2)$.

Question 16 of 22

What is $d/dx\ (x)$?

- ◯ 1 correct
- ◯ 0
- ◯ x
- ◯ x^2

Correct. By the power rule $x = x^1$, so the derivative is $1 \cdot x^0 = 1$.

Question 17 of 22

What is $d/dx\ (4x^3 - 2x^2 + 7x)$?

- ◯ $12x^2 - 4x + 7$ correct
- ◯ $12x^2 - 4x$
- ◯ $12x^3 - 4x + 7$
- ◯ $4x^2 - 2x + 7$

Correct. Term by term: $12x^2$, $-4x$, $+7$.

Question 18 of 22

Which rule differentiates a composite function `f(g(x))`?

- ◯ Chain rule — correct
- ◯ Product rule
- ◯ Quotient rule
- ◯ Sum rule

Correct. A function inside another function needs the chain rule: $f'(g(x)) \cdot g'(x)$. This is exactly what backpropagation applies layer by layer.

Question 19 of 22

Using the quotient rule, `d/dx ( x / (x + 1) )` = ?

- ◯ $1 / (x + 1)^2$ — correct
- ◯ $1 / (x + 1)$
- ◯ $x / (x + 1)^2$
- ◯ $-1 / (x + 1)^2$

Correct. Quotient rule $(u/v)' = (u'v - uv')/v^2$ with u = x, v = x+1: $(1 \cdot (x+1) - x \cdot 1)/(x+1)^2 = 1/(x+1)^2$.

Question 20 of 22

Using the chain rule, `d/dx (cos(3x))` = ?

- ◯ $-3 \cdot \sin(3x)$ — correct
- ◯ $-\sin(3x)$
- ◯ $3 \cdot \sin(3x)$
- ◯ $3 \cdot \cos(3x)$

Correct. Outer derivative $-\sin(3x)$ times inner derivative $3 \rightarrow -3 \cdot \sin(3x)$.

Question 21 of 22

Using the product rule, $d/dx\ (x \cdot e^x)$ = ?

- ◯ $e^x(x + 1)$ — correct
- ◯ e^x
- ◯ $x \cdot e^x$
- ◯ $e^x(x - 1)$

Correct. $(uv)' = u'v + uv'$ with $u = x$, $v = e^x$: $1 \cdot e^x + x \cdot e^x = e^x(x+1)$.

Question 22 of 22

What is $d/dx\ (\tan x)$?

- ◯ $\sec^2 x$ — correct
- ◯ $-\sec^2 x$
- ◯ $\sec x \cdot \tan x$
- ◯ $1/\cos x$

Correct. A standard result: the derivative of `tan x` is `sec²x` (i.e. `1/cos²x`).

Try again Scroll up to see the correct answer and a short explanation for every question.

22/22

100% correct

Flawless — you know your rules cold. 🎯

Time used: 17:16

Scroll up to review every answer · explanations appear under each question after submitting